Project Portfolio Budgeting as a Markov Decision Process: Environment Design, Baseline Benchmarks, and Optimality Gap Analysis

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Abstract

Allocating a shared capital reserve across a portfolio of simultaneously executing contracted projects is a high-frequency, high-stakes decision for engineering contractors. Classical approaches either assume full foresight or collapse the portfolio to a single project, leaving practitioners with no principled tool for the sequential, multi-project setting under real-world uncertainty. This paper closes that gap by delivering a complete, verified simulation environment that makes the problem tractable for data-driven methods.

We formalise the problem as a Markov Decision Process in which a portfolio manager allocates budget fractions across N concurrent projects at each discrete period. The state encodes Earned Value Management indicators, S-curve planned progress, milestone payment lookaheads, and a per-project breach-tolerance counter. The transition dynamics reproduce the full FIDIC contract cash-flow cycle: advance payment and recovery, progress-weighted milestone gross payments, retention withholding and release, interest charges on treasury draws, and a two-condition termination gate. The reward signal is the γ-discounted portfolio net cash flow, accumulating to an episode NPV that serves as the primary performance criterion. A full-foresight oracle bound is derived analytically, and the exact stochastic optimisation problem is shown to be intractable at portfolio scale via scenario-tree enumeration, formally motivating the MDP/RL formulation over classical stochastic programming.

The environment is released as a configurable, Gymnasium-compatible package —verified across 51 checks in 10 test sections— with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag, eliminating the black-box problem inherent to learned policies. To establish a performance baseline and motivate the need for learned agents, we run three deterministic greedy heuristics (Reach-Plan, Catchup-T, and Catchup-Next-T) and a manually computed full-information optimum across four benchmark configurations of increasing portfolio complexity. The best-heuristic optimality gap widens systematically with complexity, from 3.40 NPV units on an isolated single-project instance to 72.15 units on a three-project portfolio with overlapping cash flows — a 21-fold increase — and on the most complex configuration every greedy heuristic yields a loss while the full-information optimum remains profitable. These results confirm that standard rule-based allocation is structurally insufficient under shared-budget coordination and establish a quantified, well-defined improvement target for reinforcement learning agents. Training such agents (e.g. via PPO), evaluating them against a formal Linear Programming or

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